

Foundations: Course Intro, Rigid Body Dynamics, Quadrotor Modeling

Course notes – Autumn 2022/23

Contents

1 Chapter 0 - Course Introduction	2
1.1 Logistics	2
1.2 Syllabus (full course roadmap)	2
1.3 Weekly schedule (as planned 2022/23, useful to map chapter numbers to weeks)	3
1.4 Why UAVs / vehicle type tradeoffs	3
1.5 Generic UAV control-loop architecture (recurring diagram throughout the course)	4
1.6 Sensors and onboard architecture (preview, expanded later in Ch5/Ch6)	4
1.7 Research showcase (ISR/IDMEC lab video examples - context only, not testable content)	4
2 Chapter 1 - Rigid Body Kinematics and Dynamics	5
2.1 Reference frames and configuration	5
2.2 Rotation matrices - properties (Special Orthogonal group $SO(3)$)	5
2.3 Skew-symmetric matrix $S(\cdot)$	5
2.4 Rotation representations	6
2.5 Rigid-body kinematics	6
2.6 Rigid-body dynamics	7
2.7 Final result of Ch1 - the general 3-D rigid-body model (this is the base model reused everywhere later)	8
3 Chapter 2 - Quadrotor Dynamic Modeling	8
3.1 Actuation chain (per rotor)	8
3.2 Quadrotor input forces and moments (near-hover, negligible air velocity)	8
3.3 Full quadrotor dynamic model	9
3.4 Differential flatness of the quadrotor	9
3.5 Design considerations (SWaP tradeoffs)	10
4 Key equations reference	10
5 Open questions / unclear points	11

Source files:

- 22_23_UAVs_Ch0_Intro_Part_I.pdf (17 pages)
- 22_23_UAVs_Ch0_Intro_Part_II.pdf (40 pages)
- 22_23_UAVs_Ch1_Rigid_Body.pdf (31 pages)

- 22_23_UAVs_Ch2_Quadrotor_Modeling.pdf (30 pages)

Course: **Unmanned Aerial Vehicles / Aeronaves Robotizadas (ARob)**, MEAer, IST Lisboa, Autumn Semester 2022/2023. Faculty: José Raul Azinheira (jraz@dem.ist.utl.pt, responsible), Rita Cunha (rita@isr.tecnico.ulisboa.pt). Slides are in English; occasional Portuguese course-title text ("Aeronaves Robotizadas").

1 Chapter 0 - Course Introduction

1.1 Logistics

- Contact hours: Lectures/Exercises (T) 2h/week (Tue 9h30-11h30); Laboratory (L) 1h30/week, total 12h (Wed 10h30-12h and 12h30-14h).
- Lab: groups of 3, 2 shifts of 4 groups; 3 lab assignments (L1, L2, L3), each with 4 sessions:
 1. Modelling and identification of the drone
 2. Estimation of motion variables
 3. Motion control of the drone
- 1 report per assignment, submitted before the following lab session.
- Grading: $E = \max(E1, E2)$ (exam, 40%), $L = 0.2L1 + 0.4L2 + 0.4L3$ (labs, 60%), $F = 0.4E + 0.6L$. Exam dates 26/1/2023 and 6/2/2023 (2022/23 edition).
- Recommended bibliography:
 - Beard & McLain, *Small Unmanned Aircraft: Theory and Practice*, Princeton University Press, 2012 (ISBN 9781400840601) - this is the main reference used for Euler-angle diagrams and later sensor/estimation chapters.
 - Astrom & Murray, *Feedback Systems*, online at Caltech CDS wiki.
 - Additional: "Flying Robots" chapter in *Handbook of Robotics* (Siciliano/Khatib, 2016); *Handbook of Unmanned Aerial Vehicles* (Valavanis/Vachtsevanos, 2015); *Introduction to UAV Systems* (Fahlstrom/Gleason, 2012).

1.2 Syllabus (full course roadmap)

1. Introduction to UAVs - motivation, system architecture, UAV configuration types, applications.
2. Modelling - rigid body dynamic modelling; quadrotor dynamic modelling and design considerations.
3. Control systems design - quadrotor trajectory tracking control; root-locus/loop-shaping design; inner-outer loop control structure; linear systems and state-space models; linear state feedback + linear state observers, duality and separation principle, LQR design example.
4. Sensors for UAVs - accelerometers, rate gyros, pressure sensors, digital compasses, GNSS, proximity sensors.
5. State observers and Kalman filtering - position/velocity/attitude estimation including bias correction; connection to complementary filters.

6. Nonlinear systems - stability analysis and nonlinear control design; application to quadrotor trajectory tracking control.
7. Path following and path planning for UAVs - Voronoi graphs, RRTs, coverage algorithms, Dubins paths, waypoint transitions, optimization-based methods.

1.3 Weekly schedule (as planned 2022/23, useful to map chapter numbers to weeks)

Week	Date	Lecture	Lab
1	19-Sep	Ch0 Intro Part I	enroll
2	26-Sep	Ch0 Intro Part II	L1
3	03-Oct	Ch1 Rigid Body	
4	10-Oct	Ch2 Quadrotor Modeling	L1
5	17-Oct	Ch3 Quadrotor Control Intro	L1
6	24-Oct	Ch4 Modern Control Design	RL1
7	31-Oct	(1st November holiday)	L2
8-9	07/14-Nov	mid term	
10	21-Nov	Ch5 Sensors (Beard & McLain)	L2
11	28-Nov	Ch6 State Estimation (Beard & McLain)	L2
12	05-Dec	Q&A	RL2
13	12-Dec	Ch7 Intro Nonlinear Control	L3
14	19-Dec	Ch8 Path Following	L3
(Christmas holiday)			
15	02-Jan	Ch9 PathManager	L3
16	09-Jan	Q&A	RL3

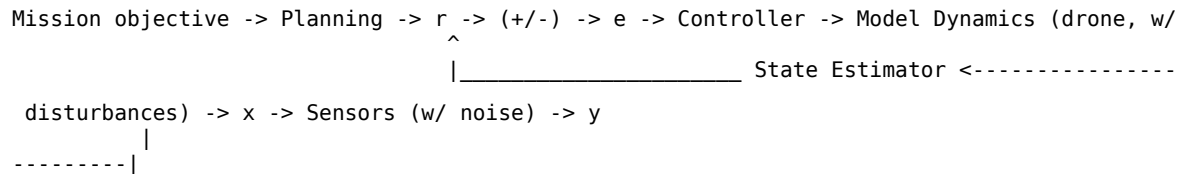
Exams: E1 26/1, E2 6/2.

1.4 Why UAVs / vehicle type tradeoffs

- Example applications: search & rescue / de-mining area survey, smuggling/forest-fire detection and tracking, parcel delivery.
- UAV must: stay airborne, navigate point-to-point, follow a road/path, avoid obstacles.
- Vehicle types compared (flying-principle comparison table, 1=bad, 3=good, criteria: power cost, control cost, payload/volume, maneuverability, stationary flight, low-speed flight, vulnerability, VTOL, endurance, miniaturization, indoor usage): Airplane (total 19), **Helicopter (25, best)**, Bird/ornithopter (24), Autogiro (18), Blimp (25, tied-best but different profile - very good hover/low power, poor payload/maneuverability).
- Quadrotor rationale: 4-DOF actuation (3-DOF omnidirectional horizontal + vertical), VTOL capable, simple/robust mechanics, good controllability, small/low-cost sensors and control hardware, but limited endurance (typically 15-30 min).
- Portuguese UAV manufacturer examples given: UAVision (fixed-wing + Spyro multicopter), Tekever (AR4 hand-launched, AR5 fixed-wing, military/defense), Spinworks (fixed-wing UAV imaging, forestry & agriculture).

- Broader application domains shown: aerial inspection, precision agriculture, entertainment, and military (ISR defense, forest-fire surveillance, maritime surveillance, search & rescue, crowd control).

1.5 Generic UAV control-loop architecture (recurring diagram throughout the course)



- General topics of the course map onto this loop: Modelling (L1), Estimation (L2), Control (L3), and Guidance/Path Planning/Trajectory generation.
- Control objective throughout: **track a trajectory**.

1.6 Sensors and onboard architecture (preview, expanded later in Ch5/Ch6)

- GPS - position estimate.
- Barometer or sonar - height estimate.
- IMU (accelerometers, gyroscopes, magnetometers) - attitude estimate.
- Cameras and LiDARs - mainly obstacle avoidance and task execution (surveying, inspection).
- Onboard computers, communication systems (LTE/WiFi/RC), batteries.
- Biggest limitation to enhanced autonomy: **flight time vs. available payload** tradeoff (empirical DJI Matrice 600 Pro curve: flight time drops from 35min at low payload to 16min at 6kg payload).
- There is a theoretical optimum: normalized flight time vs. battery-mass fraction of total mass is a concave curve peaking around battery fraction ≈ 0.65 - 0.7 (derived formally in Ch2, design considerations, where the optimum is shown to be exactly $2/3$).

1.7 Research showcase (ISR/IDMEC lab video examples - context only, not testable content)

- Stabilization with wind-disturbance rejection (fan test).
- Aggressive trajectory tracking.
- Landing on a moving platform using **vision-based control**: uses optical flow / a spherical camera projection model. Key relation: image point $p_i = P_i / \|P_i\|$, giving $P_i/d \rightarrow \chi/d$ (bearing-like measurements normalized by unknown distance d); control objective $\chi \rightarrow 0$. Optical flow field $\phi = \iint_{\mathcal{W}^2} \dot{p} dp \rightarrow \dot{\chi}/d$. Assumptions: planar target, translating platform, non-aggressive maneuvers.
- Multi-vehicle **leader-following formation control**: "trailer-like" behavior for followers. In the inertial frame the followers trace translated identical paths; in the "trailer frame" they trace different, non-overlapping paths.

- **Slung-load transportation using quadrotors:** extends the state to include load position/velocity. Quadrotor-only state $(p_Q, v_Q, R_Q, \omega_Q)$, input $(T_Q, n_Q) \in \mathbb{R}^4$, flat output (p_Q, ψ_Q) . Quadrotor+load state adds (p_L, v_L) ; flat output becomes (p_L, ψ_Q) (i.e., you now control the *load's* position, not the quadrotor's).

2 Chapter 1 - Rigid Body Kinematics and Dynamics

Goal of the chapter: derive the general 3-D rigid-body equations of motion that will later be specialized to the quadrotor.

2.1 Reference frames and configuration

- A reference frame is defined by an origin and 3 orthonormal right-handed axes.
- Two frames used throughout: inertial/world frame $\{I\}$ (flat-earth model) and body-fixed frame $\{B\}$.
- Configuration of $\{B\}$ w.r.t. $\{I\}$: $({}^I p_B, {}^I_B R) \in \mathbb{R}^3 \times SO(3)$.
 - ${}^I p_B \in \mathbb{R}^3$: position of the origin of $\{B\}$ expressed in $\{I\}$ (column vector).
 - ${}^I_B R \in SO(3)$: rotation matrix from $\{B\}$ to $\{I\}$; its columns are the coordinates of $\{B\}$'s principal axes expressed in $\{I\}$: ${}^A_B R = [{}^A x_B \ {}^A y_B \ {}^A z_B]$.
- Notation convention: leading superscript = frame the vector is expressed in; sub/superscript on R = "from $\{B\}$ to $\{A\}$ ".
- Coordinate transform: ${}^A p = {}^A_B R {}^B p$.

2.2 Rotation matrices - properties (Special Orthogonal group $SO(3)$)

- $R = [r_1 \ r_2 \ r_3] \in \mathbb{R}^{3 \times 3}$ with columns mutually orthonormal ($r_i^T r_i = 1$, $r_i^T r_j = 0$) and $\det(R) = r_1^T (r_2 \times r_3) = \pm 1$; right-handed $\Rightarrow \det(R) = 1$ ("special").
- $SO(3) = \{R \in \mathbb{R}^{3 \times 3} : RR^T = R^T R = I_3, \det(R) = 1\}$ is a **group** under matrix multiplication: closure, identity I_3 , inverse $R^{-1} = R^T$ (unique), associativity — but **not commutative**.
- Inverse/relabeling: ${}^B_A R = ({}^A_B R)^{-1} = ({}^A_B R)^T$; composition rule ${}^A_B R {}^B_C R = {}^A_C R$.
- Rotations are rigid isometries: preserve distances $\|R(p - q)\| = \|p - q\|$, preserve cross product $R(p \times q) = (Rp) \times (Rq)$, preserve inner product across frames.

2.3 Skew-symmetric matrix $S(\cdot)$

- $S(\mathbf{a})\mathbf{b} = \mathbf{a} \times \mathbf{b}$, with

$$S(\mathbf{a}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}, \quad S^T = -S.$$

- R preserves orientation: $RS(\mathbf{p}) = S(R\mathbf{p})R$ (equivalently $RS(\mathbf{p})R^T = S(R\mathbf{p})$). This identity is used repeatedly to move R across a skew term.

2.4 Rotation representations

- **Exponential map / Euler's rotation theorem**: any rotation = angle $\theta \in [0, 2\pi)$ about a unit axis $\mathbf{n}$: $R(\theta, \mathbf{n}) = e^{\theta S(\mathbf{n})} = I_3 + \theta S(\mathbf{n}) + \frac{\theta^2}{2} S(\mathbf{n})^2 + \dots$
- **Rodrigues' formula** (closed form of the series): $R(\theta, \mathbf{n}) = I_3 + \sin \theta S(\mathbf{n}) + (1 - \cos \theta) S(\mathbf{n})^2 = \mathbf{nn}^T + \sin \theta S(\mathbf{n}) - \cos \theta S(\mathbf{n})^2$. Geometric decomposition $\mathbf{p} = \mathbf{p}_{\parallel} + \mathbf{p}_{\perp}$ (parallel/perpendicular to $\mathbf{n}$) gives $R\mathbf{p} = \mathbf{p}_{\parallel} + \sin \theta S(\mathbf{n})\mathbf{p}_{\perp} + \cos \theta \mathbf{p}_{\perp}$.
- **Unit quaternions**: $q = (\cos(\theta/2), \sin(\theta/2)\mathbf{n})$ - mentioned as an alternative, singularity-free (double-cover) representation; not derived in depth here.
- **Euler angles (Z-Y-X / yaw-pitch-roll)**: any rotation decomposes into 3 elementary rotations. IST convention: $\lambda = [\phi \ \theta \ \psi]^T \in \mathbb{R}^3$ (roll ϕ , pitch θ , yaw ψ), and

$${}^I_B R = R(\lambda) = R_z(\psi)R_y(\theta)R_x(\phi).$$

Elementary matrices:

$$R_z(\psi) = \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_y(\theta) = \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix}, \quad R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{bmatrix}.$$

Intermediate frames: $\{B\} \rightarrow \{B_1\}$ roll about body x-axis; $\{B_1\} \rightarrow \{B_2\}$ pitch about $\{B_1\}$'s y-axis; $\{B_2\} \rightarrow \{B_3\} = \{I\}$ (same orientation, different origin) yaw about $\{B_2\}$'s z-axis. **Key caveat**: any 3-parameter attitude representation necessarily has singularities (gimbal lock) - stated explicitly in the slides as unavoidable for Euler angles.

2.5 Rigid-body kinematics

- Linear motion (velocity expressed in inertial frame): ${}^I \dot{\mathbf{p}}_B = {}^I \mathbf{v}_B$.
- Angular motion (angular velocity in inertial frame): ${}^I_B \dot{R} = S({}^I \omega_B) {}^I_B R$.
 - *Derivation sketch*: columns r_i of R satisfy $r_i^T r_i = 1 \Rightarrow r_i^T \dot{r}_i = 0$, so $\dot{r}_i = {}^I \omega_B \times r_i$ for some common ω (this is exactly what defines angular velocity); stacking columns gives $[\dot{r}_1 \ \dot{r}_2 \ \dot{r}_3] = S({}^I \omega_B)[r_1 \ r_2 \ r_3]$.
- Rewriting with angular velocity expressed in the **body** frame ($\omega := {}^B_I R^T {}^I \omega_B$, i.e. $\omega = R^T {}^I \omega_B$) gives the more commonly used body-rate form:

$$\dot{R} = R S(\omega).$$

Both are consistent: $RR^T = I \Rightarrow \dot{R}R^T = S({}^I \omega_B)$ (inertial-frame skew) while $R^T \dot{R} = S(\omega)$ (body-frame skew).

- **Kinematics with velocities in body frame** (the form used everywhere afterward for the quadrotor):

$$\dot{\mathbf{p}} = R\mathbf{v}, \quad \dot{R} = RS(\omega),$$

where $\mathbf{v}$ is linear velocity expressed in $\{B\}$ and ω is angular velocity expressed in $\{B\}$.

- With position also expressed in body frame ${}^B p := {}^B_I R p$: ${}^B \dot{p} = -S(\omega) {}^B p + v$.
- **Worked exercise (circular motion)**: given $p(t) = r[\cos \psi, \sin \psi, 0]^T$, $R(t) = R_z(\psi(t))$, one derives $v = R^T \dot{p} = S(\omega) p_0 = [0, r\dot{\psi}, 0]^T$ and ${}^I \omega_B = S^{-1}(\dot{R}R^T) = [0, 0, \dot{\psi}]^T$, i.e. $\omega = R\omega$ in this planar case (illustrates how to move between inertial/body representations of ω).
- **Attitude kinematics via Euler-angle rates**: $\dot{\lambda} = Q(\lambda) \omega$ where

$$Q(\lambda) = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix}.$$

Its inverse: $\omega = Q^{-1}(\lambda) \dot{\lambda} = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix} \dot{\lambda}$. **Important note explicitly called**

out: Q is *not* a rotation matrix (not orthogonal) — it's the Jacobian relating Euler-angle rates to body angular velocity, and it is singular at $\theta = \pm\pi/2$ (matches the gimbal-lock caveat above; $\cos \theta$ appears in denominators of Q).

2.6 Rigid-body dynamics

- A particle of the body at body-frame offset r : $q = p + Rr$ (inertial position), $\dot{q} = \dot{p} + RS(\omega)r$.
- **Mass** $m = \int \rho(r) dV$; **center of mass** $\bar{q} = \frac{1}{m} \int q(r) \rho(r) dV$. If $\{B\}$'s origin is placed at the center of mass, $\int r \rho(r) dV = 0$.
- **Linear momentum**: $\int \dot{q}(r) \rho(r) dV = m\dot{p}$.
- **Tensor of inertia** (about center of mass, expressed in $\{B\}$): $J = -\int \rho(r) S(r)^2 dV = \int \rho(r) (r^T r I_3 - r r^T) dV$, giving the standard symmetric matrix with entries $J_{xx} = \int \rho(r) (r_y^2 + r_z^2) dV$, off-diagonals $J_{xy} = -\int \rho(r) r_x r_y dV$, etc.
 - Angular momentum about the center of mass: $\int \rho(r) (q(r) - \bar{q}) \times \dot{q}(r) dV = R J \omega$.
 - Principal axes = eigenvectors of J ; rotating about a principal axis keeps the angular-momentum direction fixed.
 - Worked example (solid cylinder, radius r , height h): $J_{zz} = \frac{1}{2} m r^2$, $J_{xx} = J_{yy} = \frac{1}{12} m (3r^2 + h^2)$, off-diagonals zero (axes of symmetry = principal axes).
- **Newton-Euler equations of motion** (derived from conservation of linear/angular momentum in the inertial frame, then transformed into the body frame using $\dot{R} = RS(\omega)$):
 - Translational: $\frac{d}{dt}(m\dot{p}) = {}^I f \Rightarrow m\dot{v} + mS(\omega)v = f$ with $f := R^T {}^I f$ (forces expressed in body frame), i.e.

$$m\dot{v} = -S(\omega)mv + f.$$
 - Rotational: $\frac{d}{dt}(R J \omega) = {}^I \tau \Rightarrow J\dot{\omega} + S(\omega)J\omega = \tau$ with $\tau := R^T {}^I \tau$, i.e.

$$J\dot{\omega} = -S(\omega)J\omega + n \quad (n \text{ used later in place of } \tau \text{ for the applied moment}).$$

2.7 Final result of Ch1 - the general 3-D rigid-body model (this is the base model reused everywhere later)

$$\dot{p} = Rv, \quad \dot{R} = RS(\omega), \quad m\dot{v} = -S(\omega)mv + f, \quad J\dot{\omega} = -S(\omega)J\omega + n,$$

with state $X = (p, R, v, \omega)$ and input $U = (f, \tau)$ [force and moment, expressed in body frame].

3 Chapter 2 - Quadrotor Dynamic Modeling

Goal: specialize the rigid-body model of Ch1 to a quadrotor, i.e. determine the actual input u and how forces f /moments n are generated as $f = f(p, R, v, \omega, u)$, $n = n(p, R, v, \omega, u)$.

3.1 Actuation chain (per rotor)

Each rotor = electric motor + propeller: input voltage $u_i \rightarrow$ motor dynamics $\rightarrow$ spin rate $\omega_i \rightarrow$ propeller aerodynamics $\rightarrow$ thrust T_i and reaction torque Q_i .

DC motor model:

- Electrical: $L\dot{i}_i(t) = u_i(t) - Ri_i(t) - k_b\omega_i(t)$ ($k_b\omega_i$ = back-EMF, proportional to spin speed).
- Mechanical: $I\dot{\omega}_i(t) = k_t i_i(t) - \tau_i^{load}(t)$ (generated torque proportional to coil current i_i , minus load torque from the propeller).

Propeller aerodynamics (blade-element theory):

- Lift/drag at a blade element at radius r : $l(r) = \frac{\rho}{2}v_{air}^2 c c_l$, $d(r) = \frac{\rho}{2}v_{air}^2 c c_d$, with $v_{air} = [\omega_i r, v_{ind}]^T$ (rotational speed component + induced/inflow velocity), c = chord, $c_l \approx a_0 \alpha$ (lift coeff. $\approx$ linear in angle of attack α) drives thrust, c_d (drag coeff.) drives torque.
- Integrating over the blade span and summing over N_b blades gives closed forms quadratic in spin speed:

$$T_i = N_b \frac{\rho}{2} R^3 a_0 \left(\frac{\theta}{3} - \frac{v_{ind}}{2} \right) \omega_i^2 = c_T \omega_i^2, \quad Q_i = c_Q \omega_i^2.$$

- c_T, c_Q (thrust/torque constants) are typically identified experimentally via static thrust-stand tests with different payloads (example: APC propeller datasheets giving RPM $\rightarrow$ Thrust/Power/Torque/ C_P/C_T tables; non-dimensional coefficients $C_T = T/(\rho n^2 D^4)$, $C_P = P/(\rho n^3 D^5)$ as functions of advance ratio $J = V/(nD) = 2\pi V/(\omega D)$ for forward-flight/airspeed effects).
- Neglected higher-order effects mentioned: momentum-theory slipstream contraction (upstream/disk/downstream sections with $p_1 < p_2$, $v_\infty < v < v_d$), and **ground effect** at low height.

3.2 Quadrotor input forces and moments (near-hover, negligible air velocity)

- Standard + **configuration** with two pairs of counter-rotating rotors (1&3 vs 2&4, or similar) cancels reaction-torque yaw bias at equal speeds. Total thrust and body-frame moments:

$$T = \sum_i T_i, \quad n_x = l(T_2 - T_4), \quad n_y = l(T_1 - T_3), \quad n_z = \frac{c_Q}{c_T}(T_1 - T_2 + T_3 - T_4),$$

i.e. in matrix form $[T, n_x, n_y, n_z]^T = M [T_1, T_2, T_3, T_4]^T$ with the **mixer matrix**

$$M = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & l & 0 & -l \\ l & 0 & -l & 0 \\ c & -c & c & -c \end{bmatrix}, \quad c = c_Q/c_T.$$

(l = arm length; the exact row/sign pattern depends on rotor numbering/geometry — an in-class exercise asks students to redo this for the $\times$ configuration and for a hexarotor, where M becomes non-square/over-actuated: "6 inputs... but $M=?$ " is posed as an open exercise, hinting at pseudo-inverse allocation.)

- With T, n expressed in $\{B\}$ and near-hover aerodynamics negligible, plug directly into the Ch1 rigid-body dynamics:

$$f = -T e_3 + mg R^T e_3, \quad n = [n_x, n_y, n_z]^T.$$

3.3 Full quadrotor dynamic model

$\dot{p} = Rv, \quad m\dot{v} = -S(\omega)mv - T e_3 + mg R^T e_3, \quad \dot{R} = RS(\omega), \quad J\dot{\omega} = -S(\omega)J\omega + n$, equivalently in inertial-frame form $m\ddot{p} = -T R e_3 + mg e_3$. Compact form $\dot{x} = f(x, u)$ with $x = (p, v, R, \omega)$, $u = (T, n_x, n_y, n_z)$. **Underactuated system: 12 states, 4 inputs** — can only directly prescribe 4 DOF, chosen as $y = (p, \psi) \in \mathbb{R}^4$ (3D position + yaw). Block-diagram causality: $(T, n) \rightarrow$ Angular Dynamics $\rightarrow r_3$ (thrust direction unit vector) $\rightarrow$ Translation Dynamics $\rightarrow (p, v)$, with (R, ω) fed back. This is explicitly framed as *the* control-design model: no wind, no aerodynamics beyond thrust/reaction-torque, ideal actuation, valid for slow flight.

3.4 Differential flatness of the quadrotor

- A system is **flat** if state and input $(x(t), u(t))$ can be written purely as functions of a flat output $y(t)$ and a finite number of its time derivatives, and $\dim(y) = \dim(u)$.
- For the quadrotor, $y = (p, \psi) \in \mathbb{R}^4$ (matches $u = (T, n_x, n_y, n_z) \in \mathbb{R}^4$), and it can be shown $(x, u) = f(p, \dot{p}, \ddot{p}, p^{(3)}, p^{(4)}, \psi, \dot{\psi}, \ddot{\psi})$.
- **Exercise worked in class:** from $m\ddot{p} = -T R e_3 + mg e_3$, define $f_T = -T R e_3$ (thrust force) and $f_g = mg e_3$ (gravity); then $m\ddot{p} = f_T + f_g$, so T and the thrust *direction* $R e_3$ are recovered algebraically from the desired acceleration $\ddot{p}$ and yaw ψ (given ψ , roll/pitch are solved from $R_y(\theta)R_x(\phi)e_3 = [\cos \phi \sin \theta, -\sin \phi, \cos \phi \cos \theta]^T$ matched against $-\ddot{p} - f_g$ direction). This is the standard "differential flatness" trick used in most quadrotor trajectory-tracking controllers (e.g. Mellinger & Kumar-style approaches), though that paper isn't cited by name in these slides.
- Special case (hover): $\ddot{p} = 0 \Rightarrow f_T + f_g = 0$ (thrust exactly cancels gravity, vertical).
- Circular-motion exercise: given $p(t) = r[\cos \psi, \sin \psi, 0]^T$ with $\dot{\psi} = c$ constant, students derive roll ϕ and pitch θ from the same force-balance identity —

illustrating that even for a simple circular flat trajectory, non-trivial roll/pitch are required (banking into the turn).

3.5 Design considerations (SWaP tradeoffs)

- **Thrust/weight ratio:** $T_{hover} = \frac{1}{4}mg$ per rotor at hover; $T_{max} = \frac{c_T}{c_Q}Q_{max}$ (limited by max motor torque); ratio $T_{max}/T_{hover} = 4\frac{c_T}{c_Q}\frac{Q_{max}}{mg}$. Typically **not greater than 2**; lower values $\rightarrow$ slower response, less control authority. Measured examples: Intel Aero drone ratio ≈ 1.36 (arctan-shaped thrust curve vs input), Snapdragon-based platform ≈ 3.2 (much more linear thrust curve).
- **Flight time:** mechanical power per rotor $p_{motor} = \omega Q = c_Q \omega^3 = c_Q(T/c_T)^{3/2}$, i.e. **power scales as $T^{3/2}$** . Flight time $t_{flight} = \frac{E_{batt}}{4p_{motor}/\eta + p_{payload}}$ with motor efficiency $\eta \approx 70\%$; neglecting payload, $t_{flight} \propto m_{batt}/m^{3/2}$.
 - Optimal battery mass fraction $\rho = m_{batt}/m$: with $t_{flight} \propto \rho\sqrt{1-\rho}/m_0$ (where $m_0 = m - m_{batt}$ is the fixed airframe mass), the flight-time-vs-battery-fraction curve is concave and maximized at $\rho^* = 2/3$ — i.e. **about two-thirds of total mass should be battery** for max endurance. (This is the same normalized curve teased in Ch0.)
 - Battery chemistry tradeoff (specific power vs specific energy): supercapacitors/NiCd/NiMH favor high power (fast discharge, low energy density), Lithium-ion/Li-Polymer favor high energy density (longer flight time) — standard Ragone-plot style comparison.
- **Agility vs size:** characteristic length d ; mass $m \propto d^3$, inertia $J \propto d^5$. Rotor tip-speed scaling gives $\omega_i \propto 1/\sqrt{d}$ (Mach scaling) or $\omega_i \propto 1/d$ (Froude scaling); since $T \propto \omega_i^2 d^4$ and $J \propto d^5$, angular acceleration $\dot{\omega} \propto Td/J \propto \omega_i^2/d$, which reduces (with Froude scaling) to $\dot{\omega} \propto 1/d^2$ or (with the $\omega_i^2 d^4/d^5$ grouping shown) $\dot{\omega} = \omega_i^2$ scaling directly. **Conclusion stated explicitly in the slides: reducing size increases agility.** (Source cited: Mahony, Kumar, Corke, "Multirotor Aerial Vehicles," IEEE Robotics & Automation Magazine, vol 19, no 3, 2012 — a good canonical reference for this whole chapter's design-consideration section.)

4 Key equations reference

- Rotation composition / inverse: ${}^A_B R {}^B_C R = {}^A_C R$; $({}^A_B R)^{-1} = ({}^A_B R)^T$.
- Skew-symmetric cross product: $S(a)b = a \times b$; $RS(p)R^T = S(Rp)$.
- Rodrigues' formula: $R(\theta, n) = I + \sin \theta S(n) + (1 - \cos \theta)S(n)^2$.
- Euler ZYX: ${}^I_B R = R_z(\psi)R_y(\theta)R_x(\phi)$; rate map $\dot{\lambda} = Q(\lambda)\omega$ (singular at $\theta = \pm 90^\circ$).
- Rigid-body kinematics (body-frame velocities): $\dot{p} = Rv$, $\dot{R} = RS(\omega)$.
- Rigid-body dynamics (Newton-Euler, body frame): $m\dot{v} = -S(\omega)mv + f$, $J\dot{\omega} = -S(\omega)J\omega + n$.

- Quadrotor mixer: $[T, n_x, n_y, n_z]^T = M[T_1, T_2, T_3, T_4]^T$, $T_i = c_T \omega_i^2$, $Q_i = c_Q \omega_i^2 (-1)^{i+1}$.
- Quadrotor translational dynamics with thrust input: $m\ddot{p} = -TRe_3 + mge_3$.
- Underactuation: 12 states (p, v, R, ω) , 4 inputs (T, n_x, n_y, n_z) ; flat output $y = (p, \psi)$.
- Thrust/weight ratio: $T_{max}/T_{hover} = 4(c_T/c_Q)(Q_{max}/mg)$, practical max ≈ 2 .
- Power/endurance: $p_{motor} \propto T^{3/2}$; optimal battery mass fraction $\rho^* = 2/3$.
- Agility scaling: $m \propto d^3$, $J \propto d^5$; smaller $d \rightarrow$ higher angular acceleration $\rightarrow$ more agile.

5 Open questions / unclear points

- The exact sign/index convention of the mixer matrix M (which rotor pairs give n_x vs n_y) depends on a rotor-numbering diagram that differs between the "quadrotor forces and moments" slide (Ch0 Part II, p.13) and the "+ / $\times$ configuration" exercise slide (Ch2, p.13) — the course explicitly leaves the $\times$ configuration and hexarotor allocation as unsolved in-class exercises, so no canonical answer is in the slides themselves.
- The propeller aerodynamic derivation (Ch2 pp.6-7) elides the induced-velocity model v_{ind} (momentum theory) needed to get from blade-element integrals to the closed-form c_T, c_Q — likely covered by the cited textbook (Beard & McLain) rather than fully re-derived on slides.
- Quaternions are mentioned once (Ch1 p.12) as "an alternative representation" but never developed further in these four files (no quaternion kinematics/composition formulas given) — likely deferred to a lab handout or not used at all in this edition of the course.
- The DC motor's electrical time constant L/R vs mechanical time constant is not compared quantitatively; slides state the equations but do not discuss which dynamics dominate (relevant for later control design in Ch3/Ch4).